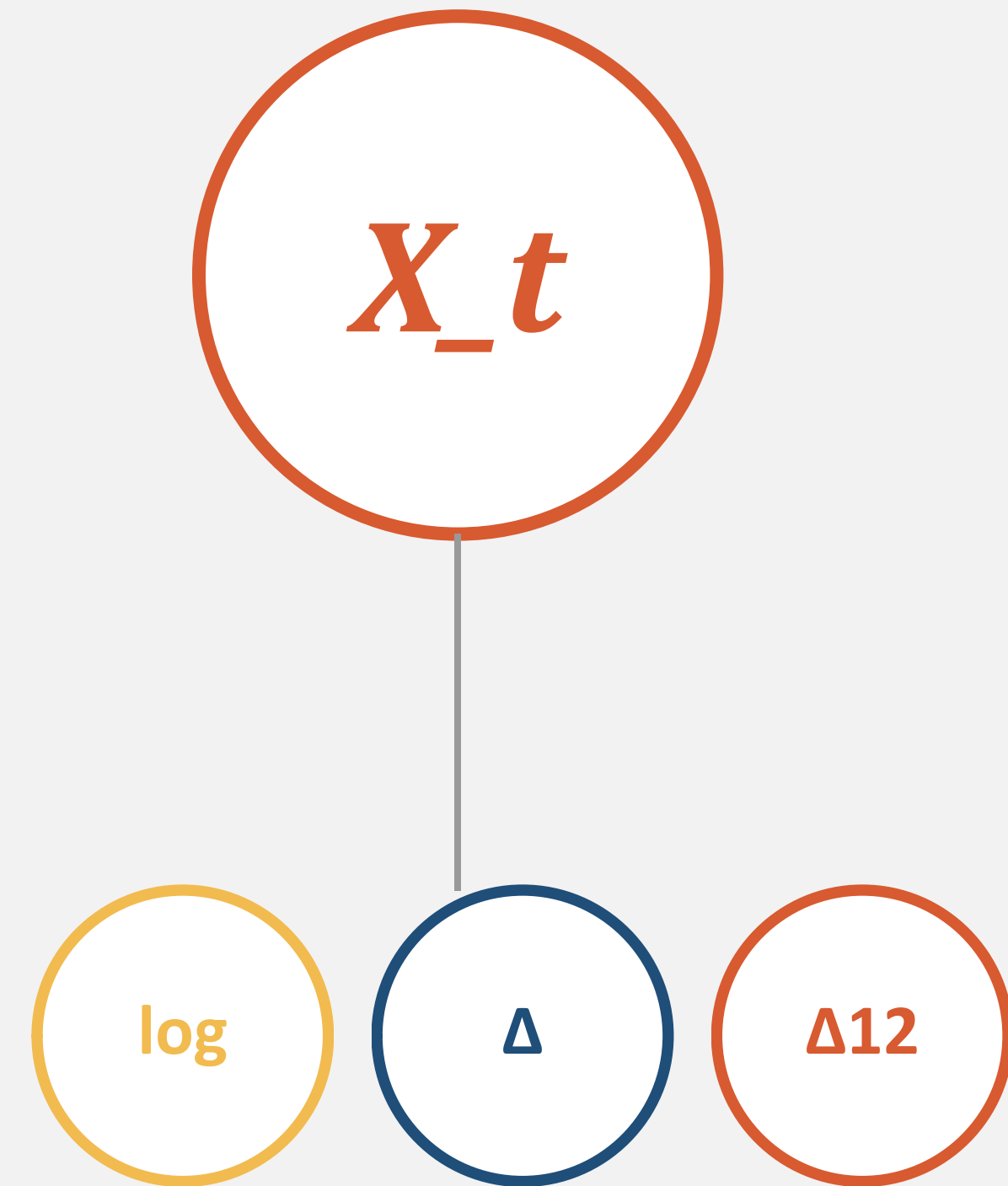


# Forecasting U.S. Food & Beverage Retail Sales

*A SARIMA Modeling Approach*

**Group 18**

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# 1. Data Introduction



## ***Data Chosen***

*U.S. Food & Beverage Stores*

**Monthly Retail Sales**

Source: U.S. Census Bureau,  
via FRED

**Series: MRTSSM445USN**

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## ***Data Preparation***

*336 monthly observations*

**Jan 1992 – Dec 2019**

Training: Jan 1992 – Dec 2018  
(324 obs)

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**Test: Jan 2019 – Dec 2019**  
**(12 obs)**

Source: <https://fred.stlouisfed.org/series/MRTSSM445USN>

## 2. Objectives & Goal



**Retail Strategy & Inventory:** Anticipate monthly demand swings to optimize stocking, staffing, and promotions.

**Supply Chain Coordination:** Give suppliers and distributors a 12-month view to align production and warehousing.

**Macroeconomic Signal:** Track household consumption a leading indicator for analysts and policymakers.

# 3. Methodologies: SARIMA

## Dataset

### U.S. Food & Beverage Stores

Monthly Retail Sales

FRED MRTSSM445USN

## Sample Period

**Training:** Jan 1992 – Dec 2018

*(324 monthly observations)*

**Test:** Jan 2019 – Dec 2019

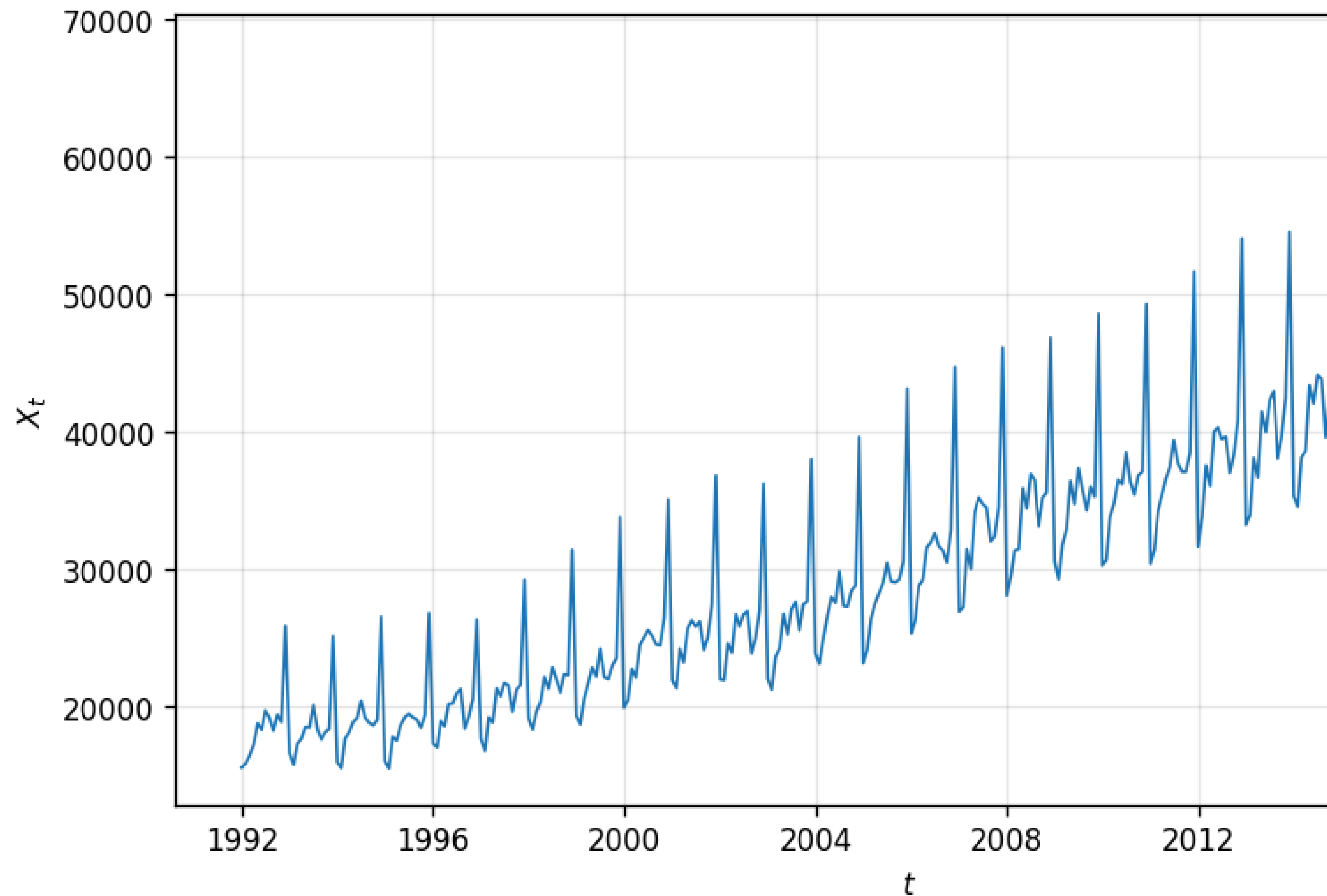
*(12 monthly observations)*

Stationarity  
Testing

Variance  
Stabilization

Differencing

### 3. Methodologies: SARIMA



$X_t$

KPSS(p-value): 0.000

**Reject  $H_0$**

*$X_t$  is not stationary*

### 3. Methodologies: SARIMA

#### *KPSS Test Summary*

**$X_t$**

*Original*

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p-value

**0.000**

**Not Stationary**

**$Y_t$**

*(1-B) $X_t$*

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p-value

**0.205**

**Stationary**

**$Z_t$**

*(1-B) $\log(X_t)$*

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p-value

**0.766**

**Most Stationary**

***Selected:** log + first differencing → cleanest stationarity result*

## 3. Methodologies: SARIMA

### Order Identification & Model Fitting

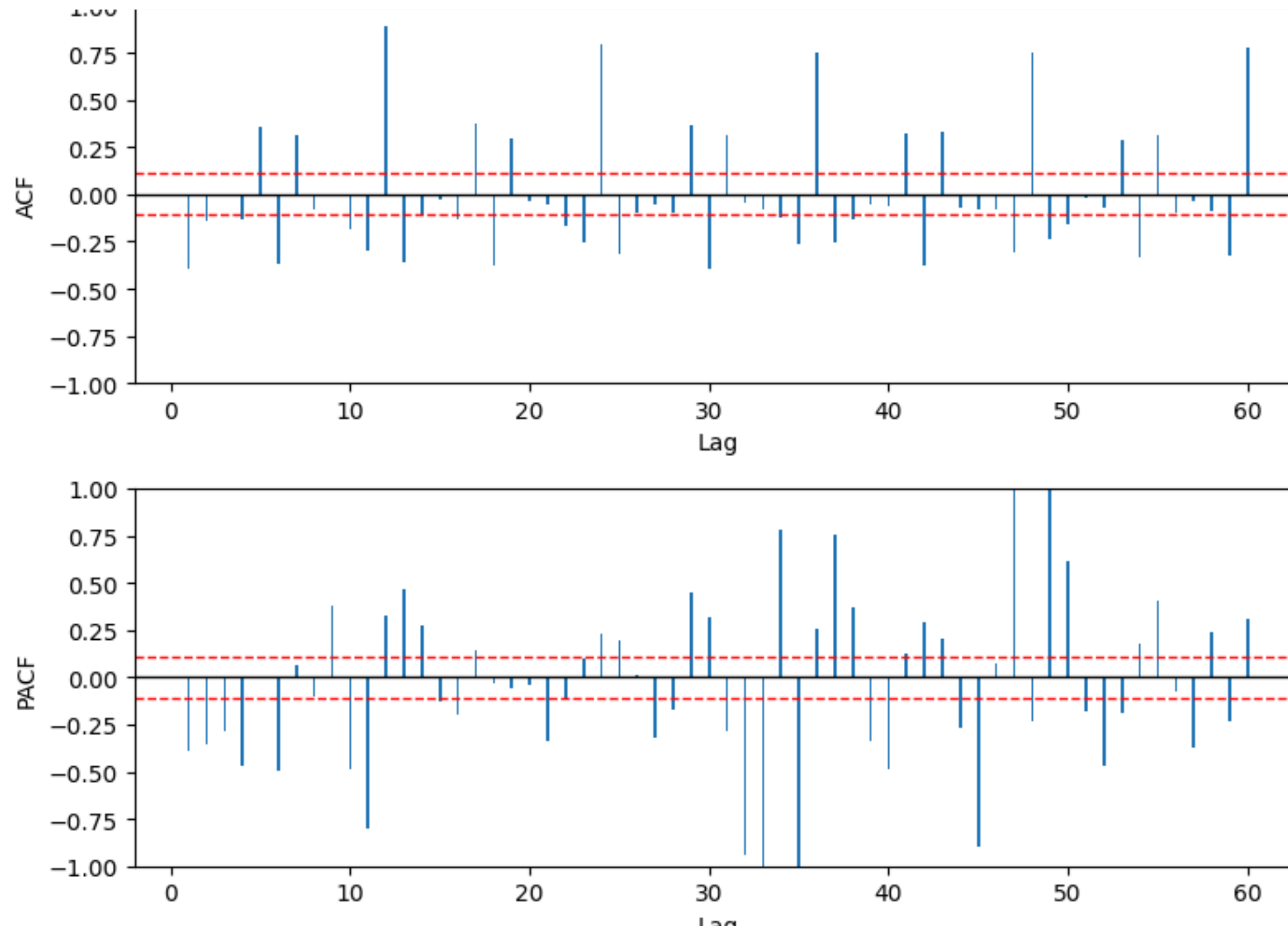
*From differenced series to a chosen SARIMA model*

ACF / PACF

Model Fitting

Selection

### 3. Methodologies: SARIMA



$$(1-B) \cdot \log(X_t)$$

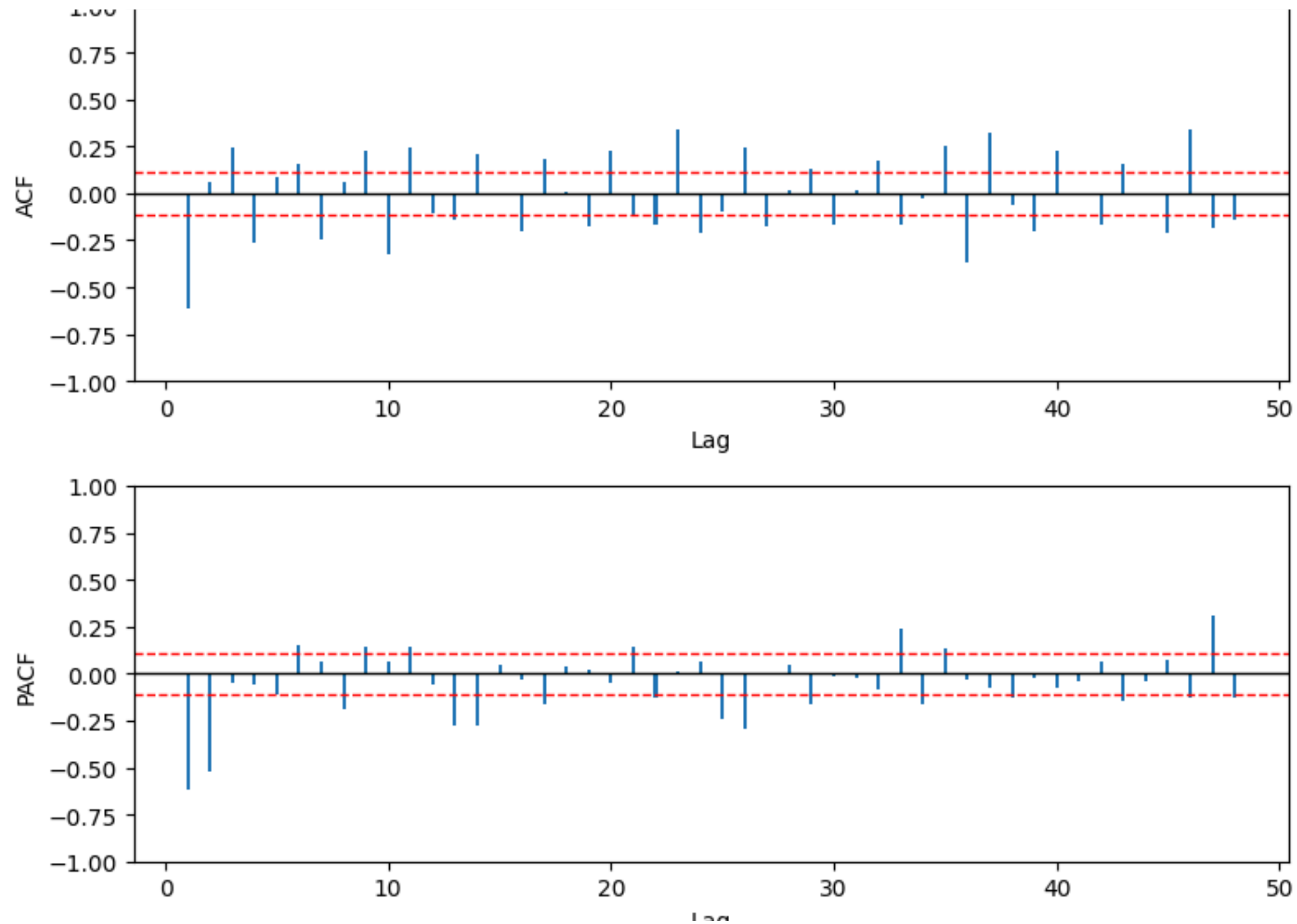
#### Observation

Strong, persistent spikes at lags 12, 24, 36, 48

#### Conclusion

Yearly cycle remains. Apply additional seasonal differencing at lag 12.

### 3. Methodologies: SARIMA



$$(1-B^{12})(1-B)\cdot\log(X_t)$$

Non-seasonal lags

**Spike at lag 1 in ACF & PACF**

$$\rightarrow p, q \in \{0, 1\}$$

Seasonal lags

**Spike at lag 12, no clear cut-off**

$$\rightarrow P, Q \in \{0, 1\}$$

### 3. Methodologies: SARIMA

#### *Candidate Model Comparison*

Model 1

$(1,1,1)(1,1,1)_{12}$

AIC

**-1755.0**

Ljung-Box p-value

**0.18**

**Strong**

Model 2

$(0,1,1)(0,1,1)_{12}$

AIC

**-1708.4**

Ljung-Box p-value

**$\approx 0.00$**

**Eliminated**

Model 3

$(1,1,1)(0,1,1)_{12}$

AIC

**-1745.2**

Ljung-Box p-value

**0.07**

**Borderline**

### 3. Methodologies: SARIMA

*Final Model Selection*

**SARIMA(1,1,1)(1,1,1)<sub>12</sub>**

*AIC*

**–1755.0**

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Lowest among candidates;  
~10 points below Model 3

*Coefficients*

**All  $p < 0.01$**

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All five terms significant  
at the 1% level

*Residual Diagnostics*

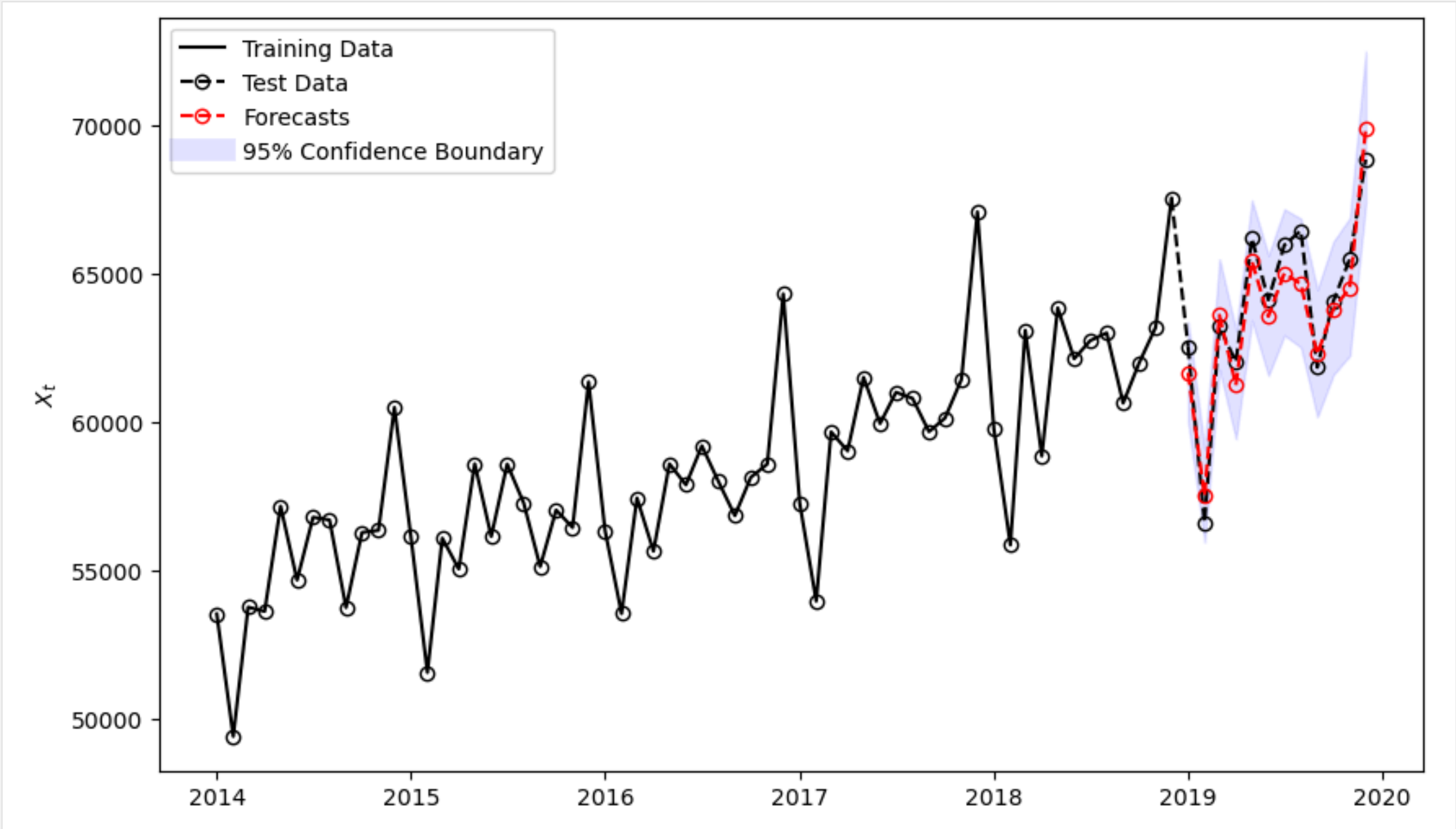
**Ljung-Box  $p = 0.18$**

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Cleanest residual structure;  
no remaining autocorrelation

*Selected on the basis of model fit and residual diagnostics — ready to forecast 2019.*

# 4. Outcomes: 2019 Forecast Results



MAPE

1.26%

Forecast Horizon

12 / 12 months

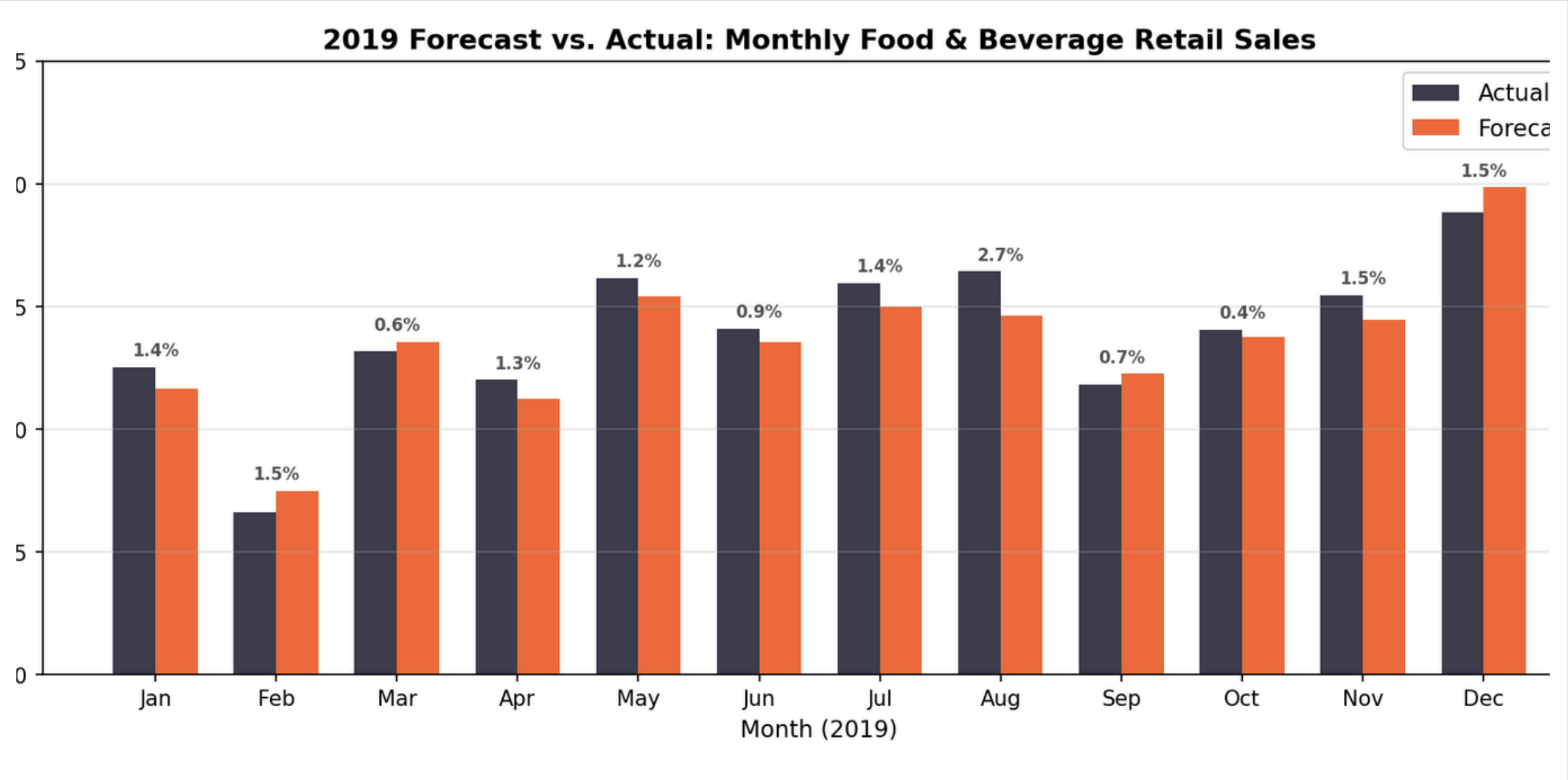
Jan 2019 – Dec 2019

Confidence Interval

95% Prediction Bounds

Model: SARIMA(1,1,1)(1,1,1)<sub>12</sub> | Fit on  $\log(X_t)$ , back-transformed to original scale

# 4. Outcomes: Accuracy & Managerial Implications



01

## Inventory Planning

Retailers can confidently plan 1-month ahead inventory orders. With MAPE < 2%, over/understock risk is minimal for food & beverage supply chains.

02

## Seasonal Staffing

The model captures Dec peak and Feb trough reliably, enabling precise seasonal workforce scheduling to match demand surges.

03

## Revenue Forecasting

SARIMA(1,1,1)(1,1,1)<sub>12</sub> provides actionable monthly revenue projections for budget allocation and financial planning cycles.

## 5. Discussion

01

### Strong Headline Performance

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**MAPE = 1.26%**  
**12 / 12 points inside 95% CI**

**Log + differencing captured trend, variance, and seasonality.**

02

### Why the Full Model Won

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**Reduced model failed Ljung-Box ( $p \approx 0.00$ ).**

**Full model cleaned residuals ( $p = 0.18$ ), dropped AIC by  $\sim 47$ .**

**Both AR and MA terms were needed.**

03

### What SARIMA Cannot See

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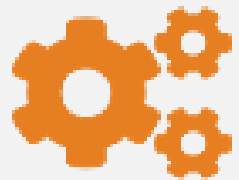
**Univariate learns only from its own history.**

**Blind to inflation, recessions, COVID.**

**A 2020 hold-out would have broken the forecast.**

## 5. Conclusion & Future Work

$\text{SARIMA}(1,1,1)(1,1,1)_{12} \rightarrow 1.26\%$  MAPE on a stable regime. 3 extensions to make it shock-robust.



01

### **SARIMAX**

Add CPI,  
unemployment,  
holidays, and income  
as regressors.



02

### **Post-COVID Stress Test**

Re-train through  
2019, forecast  
2020–2024 to  
quantify break  
impact.



03

### **Hybrid SARIMA + ML**

SARIMA handles  
seasonality; ML  
model fits residuals to  
catch non-linear  
shocks.

**Thank You!**

# References

## Data Source

[1] U.S. Census Bureau, *Retail Sales: Food and Beverage Stores [MRTSSM445USN]*, retrieved from FRED, Federal Reserve Bank of St. Louis, 2026.

<https://fred.stlouisfed.org/series/MRTSSM445USN>

## Textbook & Course Materials

[2] R. H. Shumway and D. S. Stoffer, *Time Series Analysis and Its Applications: With R Examples*, 4th ed., Springer, New York, 2017.

[3] A. Eshragh, *BU.610.740 Forecasting Models for Business Intelligence*, Lecture Notes and Computer Lab Materials, Johns Hopkins Carey Business School, Spring II, 2026.

## Methods & Software

[4] D. Kwiatkowski, P. C. B. Phillips, P. Schmidt, and Y. Shin, "Testing the null hypothesis of stationarity against the alternative of a unit root," *Journal of Econometrics*, vol. 54, pp. 159-178, 1992.

[5] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," *Proc. 9th Python in Science Conference*, 2010.